Data Flow Diagram

Date:	21 October 2026	**Team ID:**	Piyush-Balodiya-GenAI-2026
Project Name:	Multimodal Construction Intelligence Platform (CONSTRUCTAI)	**Maximum Marks:**	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system — between external entities, processes, and data stores. Use the symbol legend below as a guide.

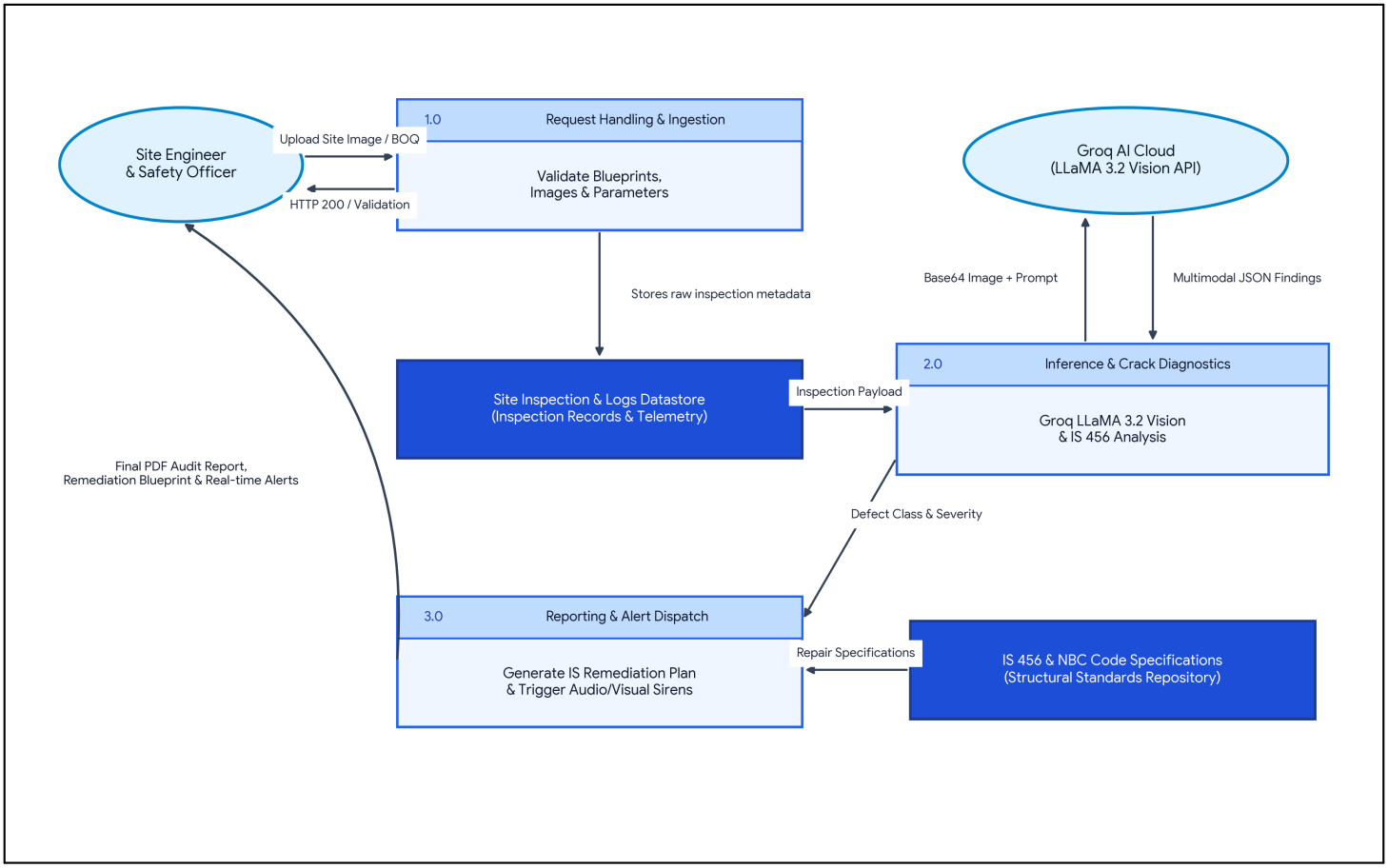
DFD Symbol Legend

Symbol	Name	Description & Project Implementation
Oval / Rounded Shape	**External Entity**	A person or external cloud system outside the application boundary that transmits or consumes data. *(e.g., Site Safety Engineer, Quality Control Officer, Groq AI LLaMA 3.2 Vision API).*
Rectangle with Numbered Header	**Process**	A discrete algorithmic activity that transforms incoming site data into outgoing insights. *(e.g., 1.0 Ingestion & Validation, 2.0 Multimodal Inference & IS 456 Analysis, 3.0 Reporting & Alert Dispatch).*
Rectangle (Solid Fill, No Number)	**Data Store**	A persistent repository where structured telemetry and safety records are archived. *(e.g., Site Inspection Records Datastore, IS 456 & NBC Regulatory Standards Base).*
Labeled Arrow	**Data Flow**	Directed trajectory of data movement between entities, processes, and data stores, labeled with explicit payloads. *(e.g., Base64 Image string, IS Code Guidelines, Inspection Report).*

Data Flow Diagram

CONSTRUCTAI System Data Flow Diagram (Level-1 Architecture)

The architectural diagram below illustrates the exact end-to-end data pipeline: from site inspection upload, FastAPI validation, Groq LLaMA 3.2 Vision multimodal reasoning, IS 456 standard correlation, to real-time audio/visual alert dispatch.



Key Data Pipeline Steps:

• **1.0 Request Handling & Ingestion:** Frontend transmits site camera snapshots or BOQ estimates via FastAPI endpoints; verifies schema & Base64 encoding.

• **2.0 AI Inference Engine:** Dispatches high-resolution imagery to Groq Cloud (LLaMA 3.2 Vision) with zero data logging for sub-second safety & crack detection.

• **3.0 Reporting & Alert Dispatch:** Correlates vision findings with IS 456 chemical repair guidelines to output executive PDF audit summaries and sound automated sirens.